NAME: RAJESH V
REG NO: 192512243
COURSE NAME : ANALOG AND DIGITAL COMMUNICATION (ECA0802)

EX NO: 2b	CALCULATION AND VERIFICATION OF FREQUENCY DEVIATION AND MODULATION INDEX OF FM SIGNALS USING MATLAB SIMULATION
DATE:	

Objectives:
To generate a Frequency Modulated (FM) signal in MATLAB using specified message and carrier signal parameters, and to determine the frequency deviation and modulation index of the FM signal while analyzing the relationship between frequency deviation, message frequency, and modulation index.

PROGRAM (MATLAB CODE)

```
% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signal
 clc;
clear;
close all;
 %% Message Signal Parameters
Am = 2;         % Message amplitude (V)
fm = 500;       % Message frequency (Hz)
 %% Carrier Signal Parameters
Ac = 1;         % Carrier amplitude (V)
fc = 10000;     % Carrier frequency (Hz)
%% FM Parameters
kf = 250;       % Frequency sensitivity (Hz/V)
fs = 100000;    % Sampling frequency (Hz)
T = 0.01;       % Simulation duration (s)
%% Time Vector
t = 0:1/fs:T;
%% Message Signal
m = Am*cos(2*pi*fm*t);
%% Frequency Deviation
delta_f = kf*Am;
 %% Modulation Index
beta = delta_f/fm;
 %% FM Signal
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
 %% Display Parameters
fprintf('Carrier Frequency (fc) = %.0f Hz\n',fc);
```

```
fprintf('Message Frequency (fm) = %.0f Hz\n',fm);
fprintf('Message Amplitude (Am) = %.0f V\n',Am);
fprintf('Frequency Sensitivity (kf) = %.0f Hz/V\n',kf);
 fprintf('\nFrequency Deviation (Delta f) = %.2f Hz\n',delta_f);
fprintf('Modulation Index (Beta) = %.2f\n',beta);
 %% Verification
beta_verify = delta_f/fm;
 fprintf('Verified Modulation Index = %.2f\n',beta_verify);
 %% Verification Check
tolerance = 1e-6;
 if abs(beta - beta_verify) < tolerance
        fprintf('\nVerification Successful\n');
else
        fprintf('\nVerification Failed\n');
end
 %% Plotting
figure;
 subplot(2,1,1);
plot(t,m,'LineWidth',1.5);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
 subplot(2,1,2);
plot(t,fm_signal,'LineWidth',1);
grid on;
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
```

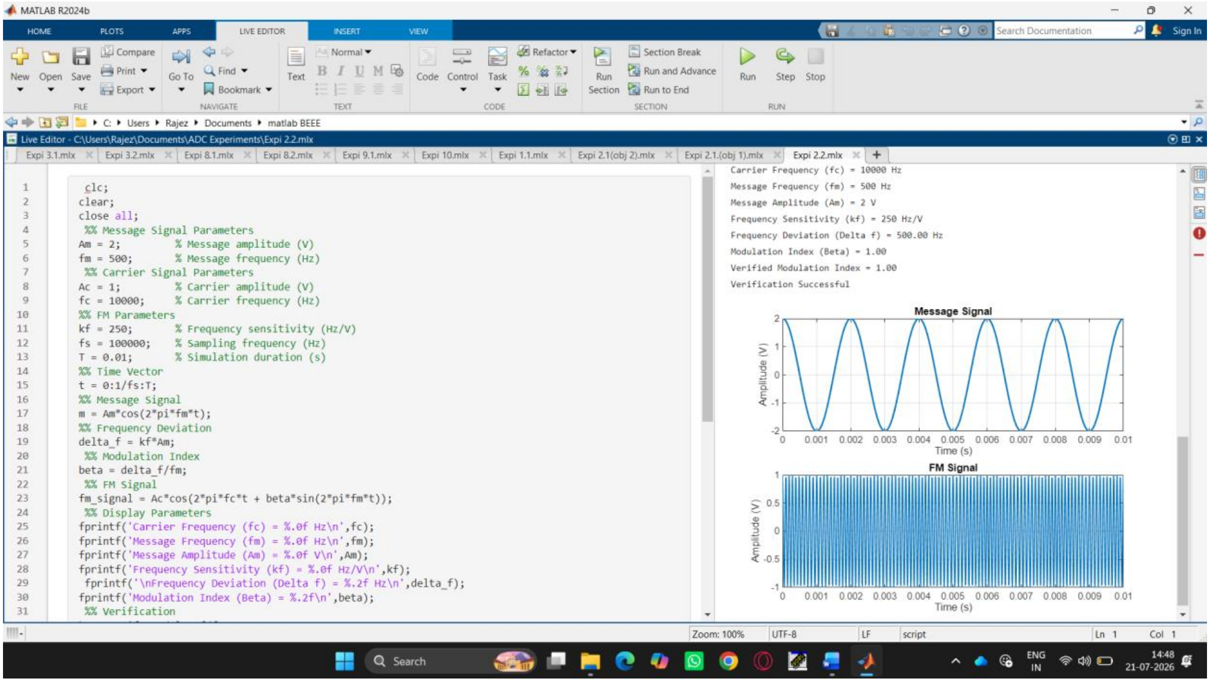
Procedure

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace variables, and close all previously opened figure windows using:
   ```
   Clc;
   Clear;
   close all;
   ```
3. Define the message signal parameters:
   - Message amplitude A_m=2V
   - Message frequency f_m=500Hz
4. Define the carrier signal parameters:
   - Carrier amplitude A_c=1V
   - Carrier frequency f_c=10000Hz

5. Define the frequency sensitivity constant k_f, sampling frequency f_s, and simulation duration T
6. Generate the time vector using:
   t = 0:1/fs:T;
7. Generate the message signal using:
   m = Am*cos(2*pi*fm*t);
   and observe the sinusoidal waveform.
8. Calculate the frequency deviation of the FM signal using:
   $\Delta f = kf.\ Am$
9. Calculate the modulation index using:
   $\beta = \Delta f / fm$
10. Generate the FM signal using:
    fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
    where the carrier frequency varies according to the message signal.
11. Plot the message signal in the first subplot and label the axes appropriately.
12. Plot the FM signal in the second subplot and observe the variation in instantaneous frequency.
13. Display the carrier frequency, message frequency, message amplitude, and frequency sensitivity in the MATLAB command window using fprintf statements.
14. Display the calculated frequency deviation and modulation index values in the command window.
15. Verify the modulation index by recomputing it using:
    $\beta = \Delta f / fm$

    and compare the result with the previously calculated value.
16. Use a conditional statement (if-else) to check whether both modulation index values are equal within a small tolerance.
17. Display the verification result as:
    - "Verification Successful" if both values match.
    - "Verification Failed" otherwise.
18. Execute the MATLAB program and observe the message signal, FM signal, calculated frequency deviation, modulation index, and verification result.

OUTPUT:



Result

This experiment verifies that the modulation index is equal to the ratio of frequency deviation to message frequency, i.e., β=Δf/f_m.